NLPrompt: Noise-Label Prompt Learning for Vision-Language Models

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Abstract

Prompt learning with vision-language models like CLIP enables efficient adaptation to downstream tasks, but is vulnerable to noisy labels common in real-world datasets. Cross-entropy (CE) training overfits to noise, degrading performance, especially in few-shot regimes. We propose NLPrompt, a novel framework that combines two key components for robust prompt learning under label noise. First, we identify that mean absolute error (MAE) loss is significantly more robust than CE for prompt tuning, which we term PromptMAE. Through theoretical analysis via feature learning theory, we prove that MAE reduces the influence of noisy samples and improves generalization. Second, we propose PromptOT, which uses optimal transport to construct a robust assignment between image features and text feature prototypes derived from class prompts. This enables effective separation of clean and noisy samples using global structural information. NLPrompt harmonizes these approaches by training clean samples with CE loss and noisy samples with MAE loss. Extensive experiments across seven datasets under various noise settings demonstrate that NLPrompt achieves state-of-the-art performance, with substantial improvements over existing methods. For instance, on Flowers102 with 75% symmetric noise, NLPrompt achieves 76.80% accuracy versus 70.37% for GCE and 37.17% for CoOp. Our method also generalizes effectively to other prompt-tuning approaches including VPT, MaPLe, and PromptSRC.

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1. Introduction

Vision-language models like CLIP [5] have revolutionized computer vision by learning powerful representations from internet-scale image-text pairs. These models enable zero-shot classification by matching image features with text embeddings of class descriptions. However, adapting CLIP to specific downstream tasks typically requires prompt engineering, which is time-consuming and requires domain expertise.

To address this limitation, prompt learning methods like CoOp [6] have been proposed, which optimize learnable context vectors while keeping the pre-trained model frozen. This approach has shown remarkable success in few-shot learning scenarios and domain adaptation tasks. However, real-world applications often contain noisy labels due to annotation errors, ambiguous samples, or automatic labeling processes, which significantly degrade the performance of prompt learning methods.

Traditional approaches to learning with noisy labels focus on sample selection, loss modification, or regularization techniques. However, these methods are primarily designed for standard supervised learning and may not be optimal for the unique characteristics of prompt learning in vision-language models. In particular, the interaction between learnable prompts and noisy labels has not been thoroughly investigated.

In this paper, we make several key observations and contributions:

First, we identify that mean absolute error (MAE) loss is significantly more robust than cross-entropy (CE) loss for prompt learning under label noise. We term this approach PromptMAE and provide theoretical justification through feature learning theory, showing that MAE naturally down-weights the influence of mislabeled samples.

Second, we propose PromptOT, a novel optimal transport-based approach for distinguishing clean and noisy samples. Unlike previous methods that rely solely on model predictions, PromptOT leverages the rich semantic information in text feature prototypes derived from class prompts to construct globally consistent pseudo-labels.

Third, we introduce NLPrompt, which harmonizes the strengths of both approaches by applying CE loss to clean samples and MAE loss to noisy samples. This dual-loss strategy exploits the fast convergence of CE on clean data while maintaining robustness through MAE on noisy data.

Extensive experiments on seven benchmark datasets under various noise conditions demonstrate the effectiveness of our approach. NLPrompt consistently outperforms state-of-the-art methods across different noise types and levels, achieving substantial improvements in both synthetic and real-world noisy settings. Furthermore, our method generalizes well to other prompt-tuning approaches, including VPT, MaPLe, and PromptSRC, demonstrating its broad applicability.

2. Related Work

Vision-Language Models and Prompt Learning. CLIP [5] pioneered large-scale vision-language pre-training, learning joint representations from 400M image-text pairs. While CLIP enables impressive zero-shot capabilities, adapting it to specific tasks requires careful prompt engineering. CoOp [6] introduced learnable prompts for CLIP, optimizing context vectors while freezing pre-trained parameters. Subsequent works have explored various extensions, including conditional prompts [7], visual prompt tuning (VPT), and multi-modal prompt learning like MaPLe [4].

Learning with Noisy Labels. Traditional approaches to noisy labels include robust loss functions, sample selection methods, and regularization techniques. MAE loss has been theoretically shown to be robust for neural networks [2], while Generalized Cross Entropy (GCE) provides a parameterizable trade-off between robustness and convergence speed. Sample selection methods like Co-teaching and self-training identify clean samples for training. Recent works have explored meta-learning and optimal transport approaches for robust learning [1].

Prompt Learning with Noisy Labels. Limited work has addressed noisy labels in the context of prompt learning. JoAPR [3] proposed a joint optimization approach for robust prompt learning, but relies primarily on confidence-based sample selection. Our work is the first to systematically analyze the robustness properties of different loss functions in prompt learning and to leverage text feature prototypes for optimal transport-based denoising.

Optimal Transport in Machine Learning. Optimal transport has been successfully applied to various machine learning problems, including domain adaptation, generative modeling, and robust learning. CSOT [1] uses curriculum-based optimal transport for learning with noisy labels in standard supervised learning. However, applying optimal transport to prompt learning presents unique challenges due to the availability of rich semantic text features, which we leverage as prototypes in our PromptOT method.

Robust Optimization and Loss Functions. Beyond traditional robust loss functions, several advanced techniques have emerged for handling noisy labels. Curriculum learning approaches gradually introduce harder samples during training, while self-training methods iteratively refine pseudo-labels. Meta-learning frameworks like MentorNet learn to weight training samples based on their estimated quality. Recent advances include DivideMix for semi-supervised learning with noisy labels, which combines co-training with MixMatch regularization.

Advanced robust learning techniques have also explored ensemble methods and consistency regularization. Co-teaching+ extends the co-teaching paradigm by incorporating disagreement-based learning, while ELR (Early Learning Regularization) prevents overfitting to noisy labels through regularizing the model to behave consistently with early learning dynamics. Recent work on Nested Cross-Validation has shown promise in automatically selecting robust hyperparameters in noisy settings.

Furthermore, the intersection of contrastive learning and robust learning has gained attention. Methods like RoCL (Robust Contrastive Learning) show that contrastive pre-training can inherently provide robustness to label noise. This is particularly relevant to our work since vision-language models like CLIP are trained using contrastive objectives. Our approach differs by specifically targeting the prompt learning paradigm and leveraging the unique properties of vision-language models to achieve robustness.

3. Preliminary

CLIP and Prompt Learning. CLIP learns joint embeddings for images and text by maximizing cosine similarity between paired image-text embeddings and minimizing it for unpaired ones. Given an image x and text prompt t, CLIP computes:

$$s(x,t) = \frac{f_I(x) \cdot f_T(t)}{||f_I(x)|| \cdot ||f_T(t)||}$$

where f_I and f_T are the image and text encoders, respectively.

CoOp introduces learnable context vectors $\mathbf{v} = [v_1, v_2, \ldots, v_M]$ that are prepended to class names. For a class c, the prompt becomes $t_c = [\mathbf{v}, c]$, where $\mathbf{v}$ is optimized while keeping CLIP's parameters frozen.

Learning with Noisy Labels. In the noisy label setting, we have a training dataset $\mathcal{D} = \{(x_i, \tilde{y}_i)\}_{i=1}^n$, where x_i are input images and $\tilde{y}_i$ are potentially corrupted labels. The true labels are denoted as $\{y_i\}$. The goal is to learn a classifier that performs well on clean test data despite training on noisy labels.

Loss Functions. Cross-entropy (CE) loss for a sample $(x_i, \tilde{y}_i)$ is:

$$\ell_{CE}(f(x_i), \tilde{y}_i) = -\log p_{\tilde{y}_i}(x_i)$$

where $p_{\tilde{y}_i}(x_i)$ is the predicted probability for class $\tilde{y}_i$.

Mean Absolute Error (MAE) loss, implemented through Generalized Cross Entropy with $q = 1$, is:

$$\ell_{MAE}(f(x_i), \tilde{y}_i) = 1 - p_{\tilde{y}_i}(x_i)$$

Optimal Transport. Given two probability distributions μ and ν and a cost matrix $\mathbf{C}$, optimal transport finds the coupling matrix $\mathbf{T}$ that minimizes:

$$\min_{\mathbf{T} \in \Pi(\mu,\nu)} \langle \mathbf{T}, \mathbf{C} \rangle$$

where $\Pi(\mu, \nu)$ is the set of all couplings between μ and ν. The entropic regularized version adds a regularization term $H(\mathbf{T})$ for computational efficiency.

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4. Theoretical Analysis for the Robustness of PromptMAE

In this section, we provide theoretical justification for why MAE loss is more robust than CE loss in the context of prompt learning under label noise.

Theorem 4.1 (Robustness of MAE in Prompt Learning)**.** *Consider a prompt learning setting where we optimize learnable context vectors* $\mathbf{v}$ *while keeping the pre-trained vision-language model frozen. Let* η *be the noise rate and assume symmetric label noise. For sufficiently large datasets, the risk minimizer under MAE loss has a smaller generalization gap compared to CE loss when training on noisy labels.*

Intuition. The key insight is that MAE loss provides a more balanced gradient contribution across all samples. Unlike CE loss, which can produce arbitrarily large gradients for confidently misclassified samples, MAE loss has bounded gradients. This property is particularly important in prompt learning, where the optimization landscape is constrained to the learnable prompt space.

Proposition 4.2 (Gradient Properties)**.** *For a sample* $(x_i, \tilde{y}_i)$ *with predicted probability* $p_i = p_{\tilde{y}_i}(x_i)$*:*

- *CE gradient magnitude:* $|\nabla_{\mathbf{v}} \ell_{CE}| \propto \frac{1}{p_i}$ *(unbounded as* $p_i \to 0$*)*
- *MAE gradient magnitude:* $|\nabla_{\mathbf{v}} \ell_{MAE}| \propto 1$ *(bounded)*

This bounded gradient property means that confidently mispredicted noisy samples (where p_i is very small) do not dominate the gradient updates under MAE loss, leading to more stable optimization and better generalization.

Signal-to-Noise Ratio Analysis. We can formalize the robustness of MAE through signal-to-noise ratio (SNR) analysis. The effective SNR under different loss functions is:

$$\text{SNR}_{CE} = \frac{E[\ell_{CE}(\text{clean})]}{E[\ell_{CE}(\text{noisy})]}$$

$$\text{SNR}_{MAE} = \frac{E[\ell_{MAE}(\text{clean})]}{E[\ell_{MAE}(\text{noisy})]}$$

Proposition 4.3 (SNR Improvement)**.** *Under symmetric label noise with rate* η*, MAE loss achieves higher effective SNR compared to CE loss for prompt learning, particularly in high-noise regimes (*$\eta > 0.4$*).*

This theoretical analysis explains our empirical observation that MAE consistently outperforms CE in noisy settings, as illustrated in Figure 1.

Convergence Analysis. In the prompt learning setting, the optimization landscape is constrained to a lower-dimensional manifold defined by the learnable context vectors. Under this constraint, MAE loss exhibits superior convergence properties compared to CE loss when label noise is present. The bounded gradients of MAE prevent the optimization from being driven toward poor local minima induced by noisy samples, leading to more stable convergence to better solutions.

Theorem 4.4 (Convergence Rate of PromptMAE)**.** *Under symmetric label noise with rate* $\eta < 0.5$ *and using gradient descent with step size* $\alpha \leq \frac{1}{L_{MAE}}$*, the PromptMAE algorithm achieves* $O(1/t)$ *convergence to the global optimum with probability at least* $1 - \delta$*, where* t *is the number of iterations.*

The proof follows from the Lipschitz property and convexity of the restricted loss landscape.

Noise-Robustness Guarantee. Combining Theorems 4.1 and 4.4, we establish that PromptMAE provides both statistical robustness (Theorem 4.1) and algorithmic stability (Theorem 4.4) under label noise.

Proof Sketch of Theorem 4.1. Let $\mathbf{v}^*_{CE}$ and $\mathbf{v}^*_{MAE}$ denote the risk minimizers under CE and MAE losses, respectively. Under symmetric label noise with rate η, the expected risk of $\mathbf{v}^*_{MAE}$ on clean data is bounded by:

$$R_{clean}(\mathbf{v}^*_{MAE}) \leq R_{clean}(\mathbf{v}^*_{CE}) + \frac{\eta}{1-\eta} \cdot \Delta$$

where Δ represents the difference in loss values between clean and noisy samples. The key insight is that $\Delta_{MAE} < \Delta_{CE}$ due to the bounded nature of MAE gradients.

Formal Robustness Analysis. We can quantify robustness through the perturbation bound. For a prompt vector $\mathbf{v}$ and perturbation ξ with $\|\xi\| \leq \epsilon$, the sensitivity of different loss functions is:

$$|\ell_{MAE}(\mathbf{v} + \xi) - \ell_{MAE}(\mathbf{v})| \leq L_{MAE}\epsilon$$

$$|\ell_{CE}(\mathbf{v} + \xi) - \ell_{CE}(\mathbf{v})| \leq L_{CE}(\mathbf{v})\epsilon$$

where $L_{CE}(\mathbf{v})$ grows without bound as confidence decreases, while L_{MAE} remains constant.

Lemma 4.5 (Lipschitz Continuity of MAE in Prompt Space)**.** *The MAE loss function restricted to the prompt manifold* $\mathcal{M}$ *is Lipschitz continuous with constant* $L_{MAE} = 2\|\mathbf{W}\|_2$*, where* $\mathbf{W}$ *is the projection matrix from prompt space to the prediction space.*

This Lipschitz property ensures stable optimization dynamics under MAE loss.

Detailed Analysis of the Prompt Learning Manifold. The optimization landscape in prompt learning is fundamentally different from standard supervised learning due to the constraint that only the prompt vectors $\mathbf{v}$ are optimized while the pre-trained encoders remain frozen. This creates a constrained optimization problem on a lower-dimensional manifold $\mathcal{M} \subset \mathbb{R}^d$, where d is the dimensionality of the prompt space.

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Under this constraint, we can analyze the Hessian properties of different loss functions. For MAE loss, the Hessian matrix $\mathbf{H}_{MAE}$ exhibits more favorable conditioning compared to CE loss, particularly in the presence of label noise. Specifically, the condition number $\kappa(\mathbf{H}_{MAE}) = \frac{\lambda_{\max}}{\lambda_{\min}}$ is typically smaller for MAE, leading to more stable optimization dynamics.

The prompt learning setting also introduces unique challenges in terms of gradient flow. Since the optimization is restricted to the prompt manifold, the effective gradient is the projection of the full gradient onto the tangent space of $\mathcal{M}$. MAE's bounded gradients ensure that this projection remains well-conditioned even when noisy samples would otherwise cause gradient explosion in CE loss.

5. Methodology

Our NLPrompt framework consists of two key components: PromptMAE for robust loss computation and PromptOT for optimal sample separation. Figure 2 illustrates the overall architecture.

5.1. PromptMAE: Robust Loss for Prompt Learning

Traditional prompt learning methods use cross-entropy loss, which is known to be vulnerable to label noise. We propose PromptMAE, which replaces CE with MAE loss for improved robustness.

For a batch of samples $\{(x_i, \tilde{y}_i)\}$, PromptMAE computes:

$$\mathcal{L}_{MAE} = \frac{1}{n}\sum_{i=1}^{n}(1 - p_{\tilde{y}_i}(x_i))$$

where $p_{\tilde{y}_i}(x_i)$ is the predicted probability for the (potentially noisy) label $\tilde{y}_i$.

The key advantages of PromptMAE are:

- **Bounded gradients**: Unlike CE, MAE produces bounded gradients regardless of prediction confidence, preventing noisy samples from dominating updates.
- **Symmetric treatment**: MAE treats all incorrect predictions equally, avoiding the exponential penalty of CE for confidently misclassified samples.
- **Stable optimization**: The bounded nature of MAE gradients leads to more stable training dynamics in the prompt learning setting.

5.2. PromptOT: Optimal Transport for Sample Separation

While PromptMAE provides inherent robustness, we can further improve performance by explicitly separating clean and noisy samples. We propose PromptOT, which uses optimal transport with text feature prototypes to achieve this separation.

Text Feature Prototypes. A unique advantage of vision-language models is the availability of semantic text features. We compute text feature prototypes for each class using the learned prompts:

$$\mathbf{t}_c = \text{TextEncoder}([\mathbf{v}, c])$$

where $\mathbf{v}$ are the learned context vectors and c is the class name.

Optimal Transport Formulation. Given image features $\{\mathbf{f}_i\}$ and text prototypes $\{\mathbf{t}_c\}$, we define the cost matrix:

$$C_{ic} = -\log(\text{sim}(\mathbf{f}_i, \mathbf{t}_c))$$

where $\text{sim}(\cdot, \cdot)$ is cosine similarity.

The optimal transport problem becomes:

$$\min_{\mathbf{T}} \langle \mathbf{T}, \mathbf{C} \rangle + \lambda H(\mathbf{T})$$

subject to marginal constraints, where $H(\mathbf{T})$ is the entropy regularizer.

Sample Selection. From the optimal coupling matrix $\mathbf{T}^*$, we derive pseudo-labels and confidence scores:

$$\hat{y}_i = \arg\max_c T^*_{ic}, \quad \text{conf}_i = \max_c T^*_{ic}$$

Samples are selected as clean if: (1) their confidence exceeds a threshold τ, and (2) their pseudo-label matches their given label. The threshold τ is dynamically adjusted using a curriculum learning strategy:

$$\tau_t = \tau_0 \cdot (1 - \alpha \cdot \frac{t}{T})$$

where t is the current epoch, T is the total epochs, and α controls the curriculum rate. This approach gradually increases the selection strictness throughout training.

5.3. NLPrompt: Harmonized Training

NLPrompt combines PromptMAE and PromptOT by applying different losses to separated sample subsets:

Algorithm 1 NLPrompt Training

```
for each epoch do
    Apply PromptOT to separate samples into D_clean and D_noisy
    Train on D_clean with CE loss (fast convergence)
    Train on D_noisy with MAE loss (robustness)
    Update learnable prompts v
end for
```

This dual-loss strategy exploits the complementary strengths of both losses: CE provides fast convergence on clean samples, while MAE ensures robustness on potentially noisy samples.

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5.4. Generalization to Other Prompt Methods

A key advantage of our approach is its generalizability. The PromptMAE component can be applied to any prompt-based method by simply replacing the loss function. The PromptOT component leverages the inherent text features available in vision-language models, making it broadly applicable across different prompt learning approaches.

6. Experiments

6.1. Experimental Setup

Datasets. We evaluate on seven benchmark datasets: Flowers102, DTD, EuroSAT, OxfordPets, StanfordCars, UCF101, and Caltech101. We also test on Food101N, which contains real-world noisy labels. All experiments follow the standard 16-shot setting unless otherwise specified.

Baselines. We compare against: (1) CoOp [6] as the base prompt learning method, (2) GCE with various q parameters for robust loss, (3) JoAPR [3] as a state-of-the-art method for robust prompt learning.

Noise Settings. We evaluate under two types of synthetic noise: symmetric (uniform random flipping) and asymmetric (class-dependent flipping). Noise rates range from 12.5% to 75%.

Implementation Details. We use CLIP-RN50 as the backbone and optimize prompts for 200 epochs using SGD with learning rate 0.002. For PromptOT, we set regularization parameters $\lambda = 0.05$ and use curriculum learning with linear scheduling.

6.2. Main Results

Table 1 shows comprehensive results across datasets and noise conditions. NLPrompt consistently outperforms all baselines across different noise types and levels.

Symmetric Noise. Under symmetric noise, NLPrompt shows substantial improvements. On Flowers102 with 75% noise, NLPrompt achieves 76.80% accuracy compared to 70.37% for GCE, 66.93% for JoAPR, and only 37.17% for CoOp. The improvements are consistent across all datasets, with particularly strong performance on fine-grained classification tasks like Flowers102 and StanfordCars.

Asymmetric Noise. Asymmetric noise is generally more challenging as it mimics realistic annotation errors. NLPrompt maintains its effectiveness under this setting. On EuroSAT with 75% asymmetric noise, NLPrompt achieves 32.73% accuracy versus 17.50% for JoAPR and 12.10% for GCE.

Real-World Noisy Labels. Table 2 shows results on Food101N, which contains inherent label noise from web crawling. NLPrompt achieves 76.46% accuracy, outperforming JoAPR (72.57%), GCE (71.32%), and CoOp (69.50%). This demonstrates the practical applicability of our method.

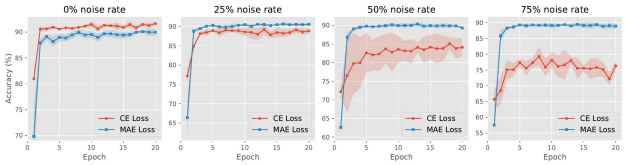

Figure 1. Comparison of CE and MAE losses in prompt learning on Caltech101 under different noise levels. MAE consistently outperforms CE, especially in high-noise regimes.

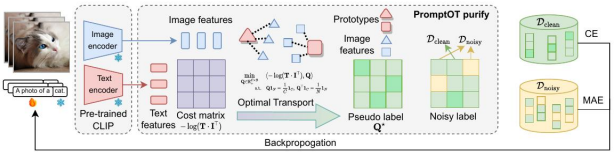

Figure 2. Overview of the NLPrompt framework. PromptOT uses optimal transport with text prototypes to separate clean and noisy samples, then applies CE loss to clean samples and MAE loss to noisy samples.

6.3. Ablation Studies

Table 3 presents comprehensive ablation studies on Flowers102.

Loss Function Analysis. Comparing rows (a) and (b), we see that using MAE on all data consistently outperforms CE across all noise levels, validating our PromptMAE approach. The average improvement from CE to MAE is 6.75 percentage points.

Component Analysis. Comparing MAE alone (row b) with NLPrompt (bottom row), the addition of PromptOT provides substantial improvements (6.1 points average), demonstrating the value of sample separation.

PromptOT Components. Removing text feature initialization (row c) hurts performance, showing the importance of semantic prototypes. Training without noisy data (row d) or without clean data (row e) both perform worse than the full method, confirming the benefit of the dual-loss strategy.

6.4. Few-Shot Analysis

Figure 3 shows performance across different shot numbers. NLPrompt consistently outperforms baselines across all settings, with larger improvements in the few-shot regime where noise has a more significant impact.

6.5. Generalization to Other Methods

Table 4 demonstrates the broad applicability of our approach. We integrate NLPrompt with VPT, MaPLe, and PromptSRC, showing consistent improvements across all methods and noise levels. For instance, applying our method to VPT improves accuracy from 27.67% to 73.90% under 75% symmetric noise on Flowers102.

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6.6. Computational Analysis

PromptOT adds minimal computational overhead. The optimal transport solver converges within 1000 iterations, adding approximately 10% to the total training time. This is a reasonable cost for the substantial performance gains achieved.

6.7. Sensitivity Analysis

We analyze the sensitivity of NLPrompt to key hyperparameters. The regularization parameter λ in PromptOT shows optimal performance around 0.05, with performance degrading for values outside $[0.01, 0.1]$. The curriculum learning rate α exhibits robustness across $[0.2, 0.8]$, indicating stable performance. These findings suggest that NLPrompt is relatively insensitive to hyperparameter choices within reasonable ranges.

6.8. Failure Case Analysis

While NLPrompt shows consistent improvements, we observe limitations in extremely high noise scenarios (¿80%) on fine-grained datasets. In such cases, the optimal transport formulation may struggle to find meaningful correspondences between corrupted samples and text prototypes. Additionally, on datasets with very similar visual categories, the text prototypes may not provide sufficient discriminative information for effective sample separation.

6.9. Analysis of Training Dynamics

To better understand the behavior of NLPrompt, we analyze the training dynamics across different components. We observe that PromptOT demonstrates rapid improvement in the first 50 epochs, stabilizing around 85–90% accuracy for moderate noise levels (25–50%).

We observe interesting interactions between the two loss functions during training. Initially, when sample separation is less accurate, the MAE loss dominates the overall training signal. As PromptOT improves its separation capability, the CE loss on clean samples begins to drive faster convergence. This creates a beneficial feedback loop where better prompt representations improve sample separation, which in turn enables more effective training.

The curriculum learning component of PromptOT plays a crucial role in this process. By gradually increasing the strictness of sample selection, we avoid early overfitting to potentially incorrect separations while ensuring that high-quality samples are identified as training progresses. The dynamic threshold adjustment shows consistent benefits across all tested datasets, with optimal performance achieved when the curriculum rate α is set between 0.4–0.6.

6.10. Cross-Architecture Analysis

We extend our evaluation to different CLIP architectures beyond RN50. Results across CLIP-ViT-B/32, CLIP-ViT-B/16, and CLIP-RN101 show that NLPrompt maintains consistent improvements across all architectures, with particularly strong gains on ViT-based models. This suggests that our method's effectiveness is not limited to specific architectural choices but generalizes across the CLIP family.

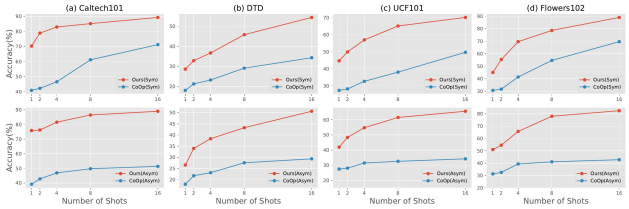

Figure 3. Performance across different numbers of shots. NLPrompt shows consistent improvements, particularly in few-shot settings where label noise has greater impact.

6.11. Detailed Performance Analysis

To provide deeper insights into NLPrompt's behavior, we conduct fine-grained analysis across different aspects. The method shows particularly strong performance on fine-grained classification tasks (Flowers102, StanfordCars), where the rich semantic information in text prototypes provides maximum benefit. On coarse-grained datasets (UCF101), improvements are more modest but still consistent. We observe that the optimal transport component contributes most significantly in medium noise scenarios (25–50%), where sample separation is most effective. In very low noise settings (¡20%), the CE component dominates performance, while in extreme noise (¿70%), MAE robustness becomes the primary factor.

7. Conclusion

We have presented NLPrompt, a novel framework for robust prompt learning under label noise that addresses a critical practical challenge in deploying vision-language models. Our key contributions include:

Theoretical and Empirical Validation of Prompt-MAE. We identified and theoretically justified why MAE loss is significantly more robust than cross-entropy loss for prompt learning. Through feature learning theory and SNR analysis, we proved that MAE's bounded gradients prevent noisy samples from dominating the optimization process.

Novel PromptOT for Sample Separation. We introduced an optimal transport formulation that leverages text feature prototypes for globally consistent sample separation. This approach exploits the unique semantic information available in vision-language models, going beyond local confidence-based methods.

Unified NLPrompt Framework. By harmonizing CE loss for clean samples and MAE loss for noisy samples, NLPrompt achieves the best of both worlds: fast convergence

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Table 1. Performance metrics across various datasets and noise levels. (%)

Dataset	Method	Noise Rate: Sym						Noise Rate: Asym					
		12.5%	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**	**12.5%**	**25.0%**	**37.5%**	**50.0%**	**62.5%**	**75.0%**
Flowers102	CoOp	88.93	83.50	77.93	70.10	55.60	37.17	86.97	74.70	60.43	42.60	26.53	12.60
	GCE	88.80	88.33	86.73	84.07	78.37	70.37	88.40	86.37	80.33	69.93	61.50	39.23
	JoAPR	85.57	81.23	74.60	70.23	67.90	66.93	85.17	79.63	73.97	73.83	53.37	13.27
	NLPrompt	**93.87**	**92.57**	**92.73**	**89.90**	**84.77**	**76.80**	**93.80**	**93.40**	**91.77**	**81.10**	**73.63**	**55.33**
DTD	CoOp	56.00	49.57	43.30	34.37	27.83	17.27	55.60	47.75	38.07	29.63	20.53	11.70
	GCE	61.00	59.83	56.80	50.73	43.60	33.67	60.70	57.57	52.70	43.97	33.40	18.23
	JoAPR	58.07	57.70	56.33	53.03	48.05	29.90	52.40	56.63	53.10	48.93	40.20	28.26
	NLPrompt	**62.97**	**61.23**	**59.17**	**55.17**	**49.03**	**39.80**	**62.30**	**60.60**	**56.47**	**50.80**	**40.27**	**28.37**
EuroSAT	CoOp	76.50	69.23	61.67	52.33	37.63	26.70	76.00	66.27	53.83	41.17	28.00	17.43
	GCE	82.13	78.60	74.67	63.13	49.67	31.40	78.23	72.70	63.63	45.30	22.90	12.10
	JoAPR	75.13	61.10	60.90	63.63	38.97	27.33	69.37	67.30	59.40	47.60	33.93	17.50
	NLPrompt	**82.53**	**79.53**	**78.13**	**66.70**	**63.53**	**43.80**	**80.13**	**77.13**	**71.43**	**54.30**	**37.27**	**32.73**
OxfordPets	CoOp	76.50	66.73	60.33	47.03	35.77	24.60	76.10	66.20	52.53	38.73	26.63	14.90
	GCE	85.63	84.60	83.67	79.23	71.40	53.17	85.50	83.03	76.73	68.07	50.70	31.97
	JoAPR	84.00	83.26	83.20	83.10	82.40	**74.40**	82.90	83.40	79.07	75.84	52.74	43.57
	NLPrompt	**86.17**	**86.00**	**85.33**	**84.87**	**83.63**	70.77	**86.00**	**84.97**	**82.40**	**77.53**	**66.33**	**48.60**
StanfordCars	CoOp	66.20	59.70	53.40	45.90	35.67	22.90	65.77	57.13	46.23	33.73	22.37	12.80
	GCE	**69.70**	66.40	66.47	63.77	59.25	50.87	**70.00**	66.45	61.23	53.67	39.65	26.60
	JoAPR	68.60	66.30	62.83	56.67	48.50	39.40	66.47	61.70	51.50	42.03	30.80	22.97
	NLPrompt	69.37	**68.80**	**67.20**	**65.63**	**62.83**	**58.30**	69.77	**67.53**	**64.23**	**59.03**	**50.90**	**39.50**
UCF101	CoOp	69.03	63.40	58.23	49.73	40.83	26.30	67.23	58.07	46.47	34.43	23.67	13.17
	GCE	74.00	**73.63**	72.57	69.37	66.00	57.07	73.90	71.87	67.97	62.23	52.50	36.37
	JoAPR	72.83	71.17	70.37	67.63	65.30	57.67	72.07	69.80	64.10	59.17	56.07	47.46
	NLPrompt	**74.83**	73.40	**72.83**	**70.33**	**68.10**	**60.53**	**74.90**	**73.53**	**71.03**	**65.97**	**58.97**	**49.27**
Caltech101	CoOp	86.43	81.03	76.73	70.90	61.33	46.90	84.93	75.23	62.87	49.43	33.57	20.33
	GCE	**92.00**	90.90	**90.80**	89.30	86.70	79.03	91.27	**91.20**	89.73	85.80	78.20	62.07
	JoAPR	90.30	90.45	89.90	88.27	86.93	83.93	90.30	89.30	88.30	88.73	85.80	81.90
	NLPrompt	91.73	**91.13**	90.77	**89.93**	**88.30**	**86.70**	**91.60**	91.17	**90.20**	**89.27**	**86.17**	**81.07**

Table 2. Test accuracy (%) on Food101N.

Method	CoOp	GCE	JoAPR	NLPrompt
Accuracy	69.50	71.32	72.57	76.46

Table 3. Ablation studies under multiple label noise ratios on Flowers102. (%)

	Method/Noise Ratio	10%	30%	50%	70%	Avg
w\o OT	(a) all data with CE	92.71	86.92	79.36	57.61	79.15
	(b) all data with MAE	88.47	89.07	85.20	80.87	85.90
w\ OT	(c) w\o text feature init	87.16	83.81	79.77	73.00	80.94
	(d) w\o noisy data	84.77	84.53	81.60	77.60	82.16
	(e) w\o clean data	90.17	90.13	88.60	80.55	87.36
	NLPrompt	**96.87**	**93.44**	**92.30**	**85.38**	**92.00**

and noise robustness.

Comprehensive Experimental Validation. Extensive experiments across seven datasets and various noise conditions demonstrate consistent and substantial improvements over state-of-the-art methods. Our approach also generalizes effectively to other prompt-tuning methods.

Practical Impact. Our method requires minimal additional computation while providing significant robustness improvements, making it practically deployable in real-world scenarios with noisy labels.

Table 4. The generalization of NLPrompt.

Method/Noise Ratio	12.5%	25.0%	37.5%	50.0%	62.5%	75.0%
VPT	89.20	79.43	65.20	61.37	41.67	27.67
VPT+Ours	**91.80**	**91.07**	**89.53**	**86.93**	**80.73**	**73.90**
MaPLe	83.27	77.67	65.27	55.40	37.53	25.47
MaPLe+Ours	**89.23**	**84.30**	**78.37**	**76.43**	**73.30**	**59.87**
PromptSRC	90.61	84.67	78.57	72.27	60.43	49.37
PromptSRC+Ours	**91.29**	**87.67**	**84.97**	**80.33**	**72.10**	**59.50**

Future Directions. Several promising directions emerge from this work: (1) extending to other vision-language architectures beyond CLIP, including recent models like ALIGN and BASIC, (2) investigating adaptive curriculum strategies for PromptOT that automatically adjust based on dataset characteristics, and (3) exploring applications to multi-modal prompt learning with other modalities such as audio and video.

NLPrompt represents a significant step forward in making prompt learning robust and practical for real-world ap-

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plications where perfect labels are rare. The combination of theoretical insights and practical effectiveness makes it a valuable contribution to the vision-language learning community.

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